

Final Exam

Name: _____

Please circle your section number!

SECTION 1

SECTION 2

SECTION 3

SECTION 4

CHENXI WU
(TR 1:00 PM)

NICLAS TECHNAU
(MWF 1:20 PM)

ANDER AGUIRRE
(TR 9:30 AM)

SARAH TAMMEN
(MWF 12:05 PM)

- There are 7 problems on the exam, some of them have multiple parts.
- Read the problems carefully and **budget your time wisely**.
- If you are running out of time, please at least say *how* you would solve a given problem, preferably using notation, even if you do not have the time to do so. This may earn you partial credit.
- No calculators or other electronic devices, please. **Turn off your phone.**
- You do not have to carry out complicated numerical computations, but you should simplify your answer if it is possible with reasonable effort. In particular, geometric series must be evaluated.
- Please present your solutions in a clear manner. **Justify your steps.** Numerical answers without explanation cannot get credit. Cross out the writing that you do not wish to be graded on.
- You can use the empty pages for additional space for writing if necessary.
- **Throughout** this exam $X, X_1, X_2, \dots$ and Y denote random variables.

Question	Points	Score
1	25	
2	20	
3	20	
4	11	
5	20	
6	19	
7	20	
Total:	135	

1. Suppose that X, Y are jointly continuous random variables with joint probability density function $f(x, y)$. Assume that $f(x, y) > 0$ for each $x, y \in \mathbb{R}$.

Give **an expression** in terms of $f(x, y)$ for each of the following quantities.

Note: Throughout this problem you do not need to and are advised not to justify your answer.

(Pay attention to the endpoints of any integration you may write down, and notice that X, Y might not be independent.)

(a) (5 points) The probability density function of Y .

(b) (5 points) The probability $P(X + Y < 5)$.

(c) (5 points) The expectation of $X^3 Y^2$.

Solution:

$$E[X^3 Y^2] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^3 y^2 f(x, y) dx dy.$$

(d) (5 points) The conditional probability density of X given $Y = y$.

Solution:

$$f_{X|Y}(x|y) = \frac{f(x, y)}{\int_{-\infty}^{\infty} f(x, y) dx}$$

(e) (5 points) $E[X^3|Y = y]$.

Solution:

$$E[X^3|Y = y] = \frac{\int_{-\infty}^{\infty} x^3 f(x, y) dx}{\int_{-\infty}^{\infty} f(x, y) dx}$$

2. Throughout this problem X, Y denote distinct (not necessarily independent) random variables. Further, $X_1, \dots, X_n$ denotes a family of independent and identically distributed continuous random variables.

Note: For part (a) you do not need to and are advised not justify your answer.

- (a) (5 points) Find the probability that $X_1 > X_2 > \dots > X_n$.

Solution: $\frac{1}{n!}$

- (b) (8 points) Show $\text{Cov}(X, -X) \leq 0$. Further, show $\text{Cov}(X, -X) = 0$ if and only if $P(X = E[X]) = 1$.

Solution: First we show $\text{Cov}(X, -X) \leq 0$. For any random variables X, Y , we know

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y).$$

Taking $Y = -X$ produces $0 = \text{Var}(0) = \text{Var}(X) + \text{Var}(-X) + 2 \text{Cov}(X, -X)$. In other words, $\text{Cov}(X, -X) = -\text{Var}(X)$. Every variance is non-negative, i.e. $\text{Var}(X) \geq 0$. So, $\text{Cov}(X, -X) \leq 0$ whence $X, -X$ are negatively correlated.

Further our computation reveals that $\text{Cov}(X, -X) = 0$ if and only if $\text{Var}(X) = 0$. We claim that $\text{Var}(X) = 0$ is equivalent to $P(X = E(X)) = 1$. Indeed, if X equals $E(X)$ with probability one then $0 \leq \text{Var}(X) = 0 = E((X - E(X))^2) \leq E(0) = 0$ yields $\text{Var}(X) = 0$. Conversely, if $\text{Var}(X) = 0$ then Chebychev's inequality yields that $X - E(X) = 0$ with probability one.

- (c) (7 points) Let X, Y be independent and identically distributed. Verify that $X + Y$ and $X - Y$ are uncorrelated.

Solution: Since X, Y are identically distributed, their moments coincide meaning that $E(X^n) = E(Y^n)$ for any $n \geq 1$. What matters here is the first moment $E(X) = \mu$, and second moment $s = E(X^2)$. To verify that $\text{Cov}(X + Y, X - Y)$ vanishes we use the linearity of the expectation to write

$$\begin{aligned} \text{Cov}(X + Y, X - Y) &= E[(X + Y - 2\mu)(X - Y - 0)] \\ &= E[(X + Y)(X - Y)] - 2\mu E[X - Y] \\ &= E[X^2] - E(Y^2) - 2\mu(\mu - \mu) \\ &= s - s - 0 = 0, \end{aligned}$$

as required.

3. This problem has two parts.

- (a) (10 points) Let $X, Y \sim \text{Unif}[-10, 10]$ be independent. Find the probability density function of $X + Y$.

Solution: Since X, Y are independent, its probability density function f_{X+Y} of $X + Y$ is given by convolving the density function $f_X(x) = \frac{1}{20}1_{[-10,10]}(x)$ with itself (because it is also the density function of Y). Thus,

$$f_{X+Y}(z) = \int_{-10}^{10} f_X(x)f_X(z-x) \, dx = \int_{-10}^{10} \frac{1}{20} \cdot 1 \cdot \frac{1}{20}1_{[-10,10]}(z-x) \, dx = \frac{1}{400} \int_{-10}^{10} 1_{[-10,10]}(z-x) \, dx.$$

To proceed we need to examine when the indicator function equals to one (which is the only relevant case since it vanishes otherwise). To this end, we observe that $-10 \leq z-x \leq 10$ is equivalent to $z-10 \leq x \leq z+10$. Because the integration variable x is already confined to the interval $[-10, 10]$, $f_{X+Y}(z) = 0$ when z is outside of the $[-20, 20]$. If $z \in [-20, 20]$, then we deduce that

$$f_{X+Y}(z) = \frac{1}{400} \int_{\max(-10, z-10)}^{\min(10, z+10)} 1 \, dx = \frac{\min(10, z+10) - \max(-10, z-10)}{400}.$$

By inspecting the cases $z \in [-20, 0]$ and $z \in [0, 20]$ separately and computing numerator, we glean

$$f_{X+Y}(z) = \begin{cases} \frac{z+20}{400}, & \text{if } -20 \leq z \leq 0, \\ \frac{20-z}{400}, & \text{if } 0 \leq z \leq 20, \\ 0, & \text{otherwise.} \end{cases}$$

(b) (10 points) Let $X \sim \text{Poisson}(2)$ and $Y \sim \text{Poisson}(7)$ be independent. Verify $X + Y \sim \text{Poisson}(9)$.

Solution: Because X, Y are independent, the probability mass function p_{X+Y} can be computed by convolving the probability mass function p_X of X with the probability mass function p_Y of Y . Hence,

$$P(X + Y = n) = \sum_{k=0}^n \left(e^{-2} \frac{2^k}{k!} \cdot e^{-7} \frac{7^{n-k}}{(n-k)!} \right) = e^{-9} \sum_{k=0}^n \frac{2^k 7^{n-k}}{k!(n-k)!}.$$

Use

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} \implies \frac{1}{k!(n-k)!} = \frac{1}{n!} \binom{n}{k}.$$

So

$$P(X + Y = n) = e^{-9} \sum_{k=0}^n \frac{2^k 7^{n-k}}{k!(n-k)!} = e^{-9} \frac{1}{n!} \sum_{k=0}^n \binom{n}{k} 2^k 7^{n-k}.$$

The binomial theorem allows us to recognise that

$$\sum_{k=0}^n \binom{n}{k} 2^k 7^{n-k} = (2 + 7)^n = 9^n.$$

Therefore $P(X + Y = n) = e^{-9} \frac{9^n}{n!}$. Thus, $X + Y$ has the Poisson distribution with parameter 9.

4. (11 points) Let X be a random variable with probability density function $f_X(x) = \frac{e^x}{e-1}$ on $[0, 1]$ and $f_X(x) = 0$ otherwise. Compute the moment generating function M_X of X .

Solution: By definition, we have that $M_X(t) = E[e^{tX}]$, provided that the expectation exists. Using the provided formula for the density gives

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} f_X(x) dx = \int_0^1 e^{tx} \left(\frac{e^x}{x-1} \right) dx = \frac{1}{e-1} \int_0^1 e^{(t+1)x} dx.$$

If $t \neq -1$, then this becomes

$$M_X(t) = \frac{1}{e-1} \left(\frac{1}{t+1} e^{(t+1)x} \Big|_{x=0}^1 \right) = \frac{1}{(e-1)(t+1)} (e^{t+1} - 1).$$

Meanwhile, for $t = -1$, we have

$$M_X(t) = \frac{1}{e-1} \int_0^1 e^{0x} dx = \frac{1}{e-1}.$$

5. A group of 40 students does a Secret Santa gift exchange, in which each student writes their name on a slip of paper, the slips are put into a bucket, and each student draws a name. Each student then buys a gift for the person whose name they drew. Let X be the number of students who drew their own name.
- (a) (7 points) Use the Principle of Inclusion and Exclusion to write a sum giving the probability that at least one person gets their own name. (There is no need to evaluate the sum.)

Solution: For $i = 1, \dots, 40$, let A_i denote the event that student i draws their own name. We are interested in computing $P(A_1 \cup \dots \cup A_{40})$. By the Principle of Inclusion and Exclusion, we have that

$$\begin{aligned} P(A_1 \cup \dots \cup A_{40}) &= \sum_{1 \leq i \leq 40} P(A_i) - \sum_{1 \leq i < j \leq 40} P(A_i A_j) + \sum_{1 \leq i < j < k \leq 40} P(A_i A_j A_k) - \dots - A_1 A_2 \dots A_{40} \\ &= \sum_{k=1}^{40} (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq 40} P(A_{i_1} \dots A_{i_k}) = \sum_{k=1}^{40} (-1)^{k+1} \binom{40}{k} \frac{(40-k)!}{40!}. \end{aligned}$$

One may further simplify to

$$P(A_1 \cup \dots \cup A_{40}) = \sum_{k=1}^{40} (-1)^{k+1} \frac{1}{k!}.$$

- (b) (5 points) Compute the expected value of X by writing X as a sum of indicator random variables.

Solution: For $i = 1, \dots, 40$, let

$$X_i = \begin{cases} 1, & \text{person } i \text{ draws their own name;} \\ 0, & \text{otherwise.} \end{cases}$$

As in part (a), let A_i be the event that person i draws their own name. The variables $X_1, \dots, X_n$ are exchangeable because they result from sampling without replacement. Thus, by linearity and exchangeability, we have

$$E[X] = E[X_1 + \dots + X_{40}] = E[X_1] + \dots + E[X_{40}] = 40E[X_1] = 40P(A_1) = 40 \left(\frac{1}{40} \right) = 1.$$

(c) (8 points) Compute the variance of X .

Solution: Let $A_1, \dots, A_{40}$ and $X_1, \dots, X_{40}$ be as in parts (a) and (b). Since $X = X_1 + \dots + X_{40}$, we have that

$$\text{Var}(X) = \sum_{1 \leq i \leq 40} \text{Var}(X_i) + 2 \sum_{1 \leq i < j \leq 40} \text{Cov}(X_i, X_j).$$

Using exchangeability, along with the fact that $\text{Var}(X_i) = \frac{1}{40} \left(\frac{39}{40} \right)$ for each i , we obtain

$$\begin{aligned} \text{Var}(X) &= 40\text{Var}(X_1) + 40(39) \text{Cov}(X_1, X_2) = 40 \left(\frac{1}{40} \right) \left(\frac{39}{40} \right) + 40(39) \text{Cov}(X_1, X_2) \\ &= \frac{39}{40} + 40(39) \text{Cov}(X_1, X_2). \end{aligned}$$

We compute $\text{Cov}(X_1, X_2)$ as follows:

$$\text{Cov}(X_1, X_2) = E[X_1 X_2] - E[X_1]E[X_2] = P(A_1 A_2) - \left(\frac{1}{40} \right)^2 = \frac{1}{40} \left(\frac{1}{39} \right) - \frac{1}{40^2}.$$

Substituting this result above, we conclude that

$$\text{Var}(X) = \frac{39}{40} + 40(39) \left(\frac{1}{40} \left(\frac{1}{39} \right) - \frac{1}{40^2} \right) = \frac{39}{40} + 1 - \frac{39}{40} = 1.$$

6. Let $X_1, X_2, \dots$ be independent and identically distributed random variables. Suppose $X_1 \sim \text{Unif}([-1, 1])$. Define

$$Y_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i.$$

In some parts of this problem, you may wish to use the fact that if $U \sim \text{Unif}([a, b])$, then $\text{Var}(U) = \frac{1}{12}(b-a)^2$.

- (a) (7 points) Use Chebyshev's inequality to find an upper bound of the probability that $Y_n > 1$.

Solution: Since $E[Y_n] = 0$, we have that

$$P(Y_n > 1) \leq P(|Y_n - 0| \geq 1) \leq \frac{\text{Var}(Y_n)}{1^2}.$$

Since $X_1, X_2, X_3, \dots$ are independent, we have that

$$\text{Var}(Y_n) = \left(\frac{1}{\sqrt{n}}\right)^2 \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n} \left(n \left(\frac{1}{2} \cdot 2^2 \right) \right) = \frac{1}{3}.$$

Substituting above, we obtain

$$P(Y_n > 1) \leq \frac{1}{3}.$$

- (b) (9 points) What is the Central Limit Theorem estimate for this probability as $n \rightarrow \infty$?

Solution: Using our computation of $\text{Var}(Y_n)$ from above, we obtain

$$\begin{aligned} P(Y_n > 1) &= P\left(\frac{Y_n - E[Y_n]}{\sqrt{\text{Var}(Y_n)}} > \frac{1 - E[Y_n]}{\sqrt{\text{Var}(Y_n)}}\right) \\ &= P\left(\frac{Y_n - E[Y_n]}{\sqrt{\text{Var}(Y_n)}} > \frac{1}{\sqrt{1/3}}\right) \approx 1 - \Phi(\sqrt{3}). \end{aligned}$$

- (c) (3 points) A non-negative random variable has $E[X] = 2$, $E[X^3] = 9$ and $E[X^4] = 20$. With this information what is the best bound you can give on $P(X \geq 5)$?

Solution: For any $k \in \mathbb{N}$, we have the bound

$$P(X \geq 5) \leq \frac{E[X^k]}{5^k}.$$

Substituting $k = 1$, $k = 3$, and $k = 4$, we obtain

$$P(X \geq 5) \leq \min \left\{ \frac{2}{5}, \frac{9}{125}, \frac{20}{625} \right\} = \min \left\{ \frac{50}{125}, \frac{9}{125}, \frac{4}{125} \right\} = \frac{4}{125}.$$

7. Let $X, Y \sim \text{Exponential}(\mu)$ be independent random variables. Let $Z = X + Y$.

(a) (10 points) Find the conditional density function of X given $Z = z$.

Solution: Recall that by the substitution rule in multivariable calculus, given $A \subseteq \mathbb{R}^2$, let $\Phi : (x, y) \mapsto (x, x + y)$, we have

$$\begin{aligned} \int_A f(x, z - x) dx dz &= \int_A f \circ \Phi^{-1}(x, z) dx dz \\ &= \int_{\Phi^{-1}(A)} f \circ \Phi^{-1} \circ \Phi(x, y) \cdot |\det(D_\Phi)| dx dy \\ &= \int_{\Phi^{-1}(A)} f(x, y) dx dy \end{aligned}$$

So the joint pdf of X and Z is

$$\begin{aligned} f_{X,Z}(x, z) &= f_{X,Y}(x, z - x) \\ &= f_X(x)f_Y(z - x) = \begin{cases} \mu^2 e^{-\mu z} & 0 < x < z \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

Alternatively, we can get joint pdf by taking derivatives of the joint cdf.

Now

$$\begin{aligned} f_Z(z) &= \int_{-\infty}^{\infty} f_{X,Z}(x, z) dx = \int_0^z \mu^2 e^{-\mu z} dx = \mu^2 z e^{-\mu z} \\ f_{X|Z}(x|z) &= \frac{f_{X,Z}(x, z)}{f_Z(z)} = \begin{cases} 1/z & 0 < x < z \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

where $z > 0$

(b) (10 points) Find the conditional expectation $E[X|Z = z]$.

Solution: The conditional distribution of X when $Z = z, z > 0$, is $\text{Uniform}((0, z))$, so $E[X|Z = z] = \frac{z}{2}$.

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Table of common probability distributions

Discrete distributions.

Name	parameters	probability mass function	expectation	variance
Bernoulli	$0 \leq p \leq 1$	$p_X(0) = 1 - p, \quad p_X(1) = p$	p	$p(1 - p)$
Binomial	$0 \leq p \leq 1$ $n \geq 1$	$p_X(k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad 0 \leq k \leq n$	np	$np(1 - p)$
Geometric	$0 < p \leq 1$	$p_X(k) = p(1 - p)^{k-1}, \quad k = 1, 2, \dots$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Poisson	$\lambda > 0$	$p_X(k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, 2, \dots$	λ	λ
Hypergeometric	$N \geq 1$ $N_A \geq 0$ $n \geq 1$	$p_X(k) = \frac{\binom{N_A}{k} \binom{N-N_A}{n-k}}{\binom{N}{n}}, \quad 0 \leq k \leq n$	$\frac{n N_A}{N}$	$\frac{N-n}{N-1} n \frac{N_A(N-N_A)}{N^2}$
Negative binomial	$k \geq 1$ $0 < p \leq 1$	$p_X(n) = \binom{n-1}{k-1} p^k (1 - p)^{n-k}, \quad n \leq k$	$\frac{k}{p}$	$k \frac{1-p}{p^2}$

Continuous distributions.

Name	parameters	probability density function	expectation	variance
Uniform	$a < b$	$f_X(t) = \frac{1}{b-a}$ for $t \in [a, b]$	$\frac{a+b}{2}$	$\frac{1}{12}(b-a)^2$
Normal	μ real, $\sigma^2 > 0$	$f_X(t) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(t-\mu)^2}{2\sigma^2}}$	μ	σ^2
Exponential	$\lambda > 0$	$f_X(t) = \lambda e^{-\lambda t}$ for $t \geq 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Gamma	$r \geq 1, \quad \lambda > 0$	$f_X(t) = \frac{\lambda^r t^{r-1}}{\Gamma(r)} e^{-\lambda t}$ for $t \geq 0$	$\frac{r}{\lambda}$	$\frac{r}{\lambda^2}$
Beta	$r, s > 0$	$f_X(t) = \frac{\Gamma(r+s)}{\Gamma(r)\Gamma(s)} t^{r-1} (1-t)^{s-1}$ for $0 \leq t \leq 1$	$\frac{r}{r+s}$	$\frac{rs}{(r+s)^2(r+s+1)}$

The following table shows approximate values of the cumulative distribution function of the standard normal distribution.

x	$\Phi(x)$
-2.0	0.022
-1.9	0.028
-1.8	0.035
-1.7	0.044
-1.6	0.054
-1.5	0.066
-1.4	0.080
-1.3	0.096
-1.2	0.115
-1.1	0.135
-1.0	0.158
-0.9	0.184
-0.8	0.211
-0.7	0.242
-0.6	0.274
-0.5	0.308
-0.4	0.344
-0.3	0.382
-0.2	0.420
-0.1	0.460
0.0	0.500
0.1	0.539
0.2	0.579
0.3	0.617
0.4	0.655
0.5	0.691
0.6	0.725
0.7	0.758
0.8	0.788
0.9	0.815
1.0	0.841
1.1	0.864
1.2	0.884
1.3	0.903
1.4	0.919
1.5	0.933
1.6	0.945
1.7	0.955
1.8	0.964
1.9	0.971
2.0	0.977